

Value Iteration

Algo (Value Iteration)

$$V_{k+1} = f(V_k) = \max_{\pi} (r_{\pi} + \gamma P_{\pi} V_k), \quad k = 1, 2, 3, \dots$$

STEP1 (policy update):

$$\pi_{k+1} = \arg \max_{\pi} (r_{\pi} + \gamma P_{\pi} V_k)$$

practical elementwise form:

$$\pi_{k+1}(s) = \arg \max_{\pi} \sum_a \pi(a|s) \left(\sum_r p(r|s, a) r + \gamma \sum_{s'} p(s'|s, a) V_k(s') \right), \quad s \in S$$

let

$$Q_k^*(s) = \arg \max_a Q_k(a, s).$$

 $\Rightarrow$

$$\pi_{k+1}(a|s) = \begin{cases} 1, & a = Q_k^*(s) \\ 0, & a \neq Q_k^*(s) \end{cases} \quad \text{greedy \& deterministic}$$

STEP2 (Value update):

$$V_{k+1} = r_{\pi_{k+1}} + \gamma P_{\pi_{k+1}} V_k.$$

practical elementwise form:

$$V_{k+1}(s) = \sum_a \pi(a|s) \left(\sum_r p(r|s, a) r + \gamma \sum_{s'} p(s'|s, a) V_k(s') \right), \quad s \in S$$

 $\Rightarrow$

$$V_{k+1}(s) = \max_a Q_k(a, s)$$

Whole Process: $V_k(s) \rightarrow Q_k(s, a) \rightarrow \pi_{k+1}(a|s) \rightarrow V_{k+1} = \max_a Q_k(s, a)$
greedy

App: use Q -table

Policy Iteration

Algo (Policy Iteration)

Init: π_0

STEP1 policy eval (PE):

$$V_{\pi_k} = r_{\pi_k} + \gamma P_{\pi_k} V_{\pi_k} \Rightarrow V_{\pi_k}$$

practical elementwise form:

$$V_{\pi_k}^{(j+1)}(s) = \sum_a \pi(a|s) \left(\sum_r p(r|s, a) r + \gamma \sum_{s'} p(s'|s, a) V_{\pi_k}^{(j)}(s') \right), \quad s \in S.$$

stop con: $j \rightarrow \infty$ or $\|V_{\pi_k}^{(j+1)} - V_{\pi_k}^{(j)}\| < \epsilon$.

STEP 2 policy impro (PI):

$$\pi_{k+1} = \operatorname{argmax}_{\pi} (r_{\pi} + \gamma P_{\pi} V_{\pi_k})$$

practical elementwise form:

$$V_{\pi_k}(s') \rightarrow Q_{\pi_k}(s, a)$$

let

$$a_k^*(s) = \operatorname{argmax}_a Q_{\pi_k}(a, s)$$

then

$$\pi_{k+1}(a|s) = \begin{cases} 1, & a = a_k^*(s) \\ 0, & a \neq a_k^*(s) \end{cases}$$

whole process: $\pi_0 \xrightarrow{PE} V_{\pi_0} \xrightarrow{PI} \pi_1 \xrightarrow{PE} \dots$

Truncated policy iteration = policy iteration + j_{truncate} PE

$$\begin{cases} j_{\text{truncate}} = 1 \Rightarrow \text{value iteration} \text{ ①} \\ j_{\text{truncate}} = \infty \Rightarrow \text{policy iteration} \text{ ②} \end{cases}$$